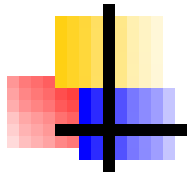


# Bacterial Metabolism

- **Metabolism**; all the biochemical rxn that occur in a cell or organism
- In prokaryotic cells is more active, versatile, and diverse than that of human cells
- Bacteria **must obtain or synthesize elements used as** building blocks of the cell.
- Microbial metabolism involves the biochemical reactions
  - to break down organic compounds
  - to synthesize new bacterial parts
- Biochemical reactions in the cell **depends** on the presence and activity of specific **enzymes**.
- Thus metabolism can be regulated
  - regulating the production of an enzyme itself
  - by regulating the activity of the enzyme

# Metabolism

- two major parts.
  - **Catabolism:** reactions that “break down” molecules, generally **energy releasing** or **exergonic**
  - **Anabolism:** reactions that build new molecules, generally energy requiring or **endergonic**



exergonic reactions provide energy for endergonic ones

- From a nutritional, or metabolic viewpoint, bacteria exist in three major physiologic types and use one of three sources of energy.
  - Phototrophs(photosynthetic); sun
  - **Chemoorganotrophs(heterotrophs)**; organic molecules
  - Chemolithotrophs(autotrophs); inorganic nutrients,

## Catabolism

Energy-yielding metabolism

Energy sources

Heat

Metabolic products

Utilizable  
energy

ATP

ADP

## Anabolism

Biosynthetic metabolism

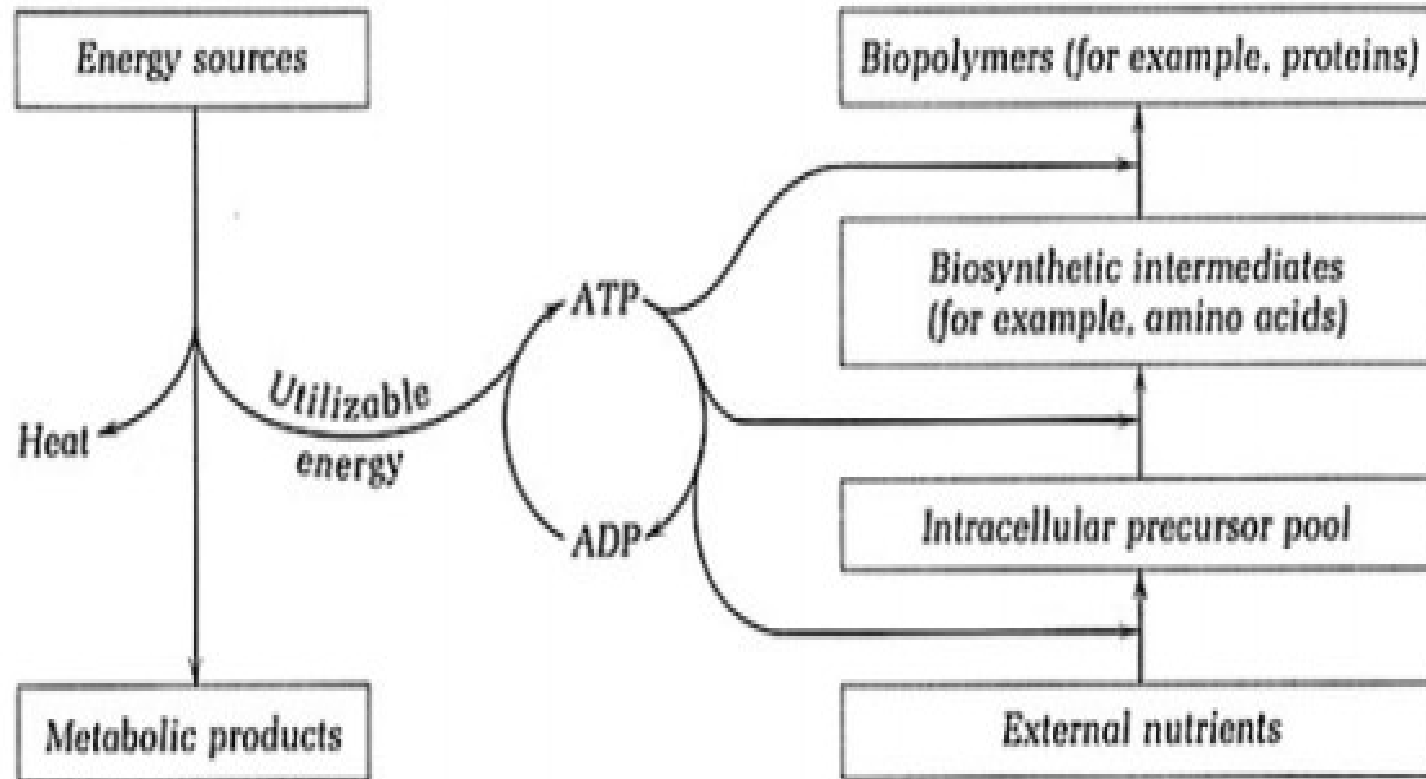
Biopolymers (for example, proteins)

Biosynthetic intermediates  
(for example, amino acids)

Intracellular precursor pool

External nutrients

The relationship between catabolism and anabolism



# Heterotrophic metabolism

- Heterotrophic metabolism is the **biologic oxidation** of organic compounds to yield ATP and simpler organic (inorganic) compounds
- Carbohydrates, lipids, and protein are the most commonly oxidized compounds.
- Oxidation of these organic compounds by bacteria results in synthesis of ATP as the chemical energy source.
- There are two common examples
  - **Respiration** yield, 38 moles of ATP
  - **Fermentation**, less energy generated
- Heterotrophic bacteria, include all pathogenic bacteria,
  - Most commonly studied bacteria;
  - They grow readily in media containing carbohydrates, proteins, or other complex nutrients such as blood.

# Autotrophic Metabolism

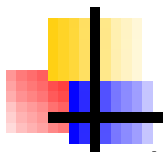
- A unique form of metabolism found only in bacteria
- Inorganic compounds are oxidized directly (without using sunlight) to yield energy (e.g.,  $\text{NH}_3$ ,  $\text{NO}_2^-$ ,  $\text{S}_2$ , and  $\text{Fe}^{2+}$ ).
- No lipid-mediated processes are involved.
- All autotrophs use  $\text{CO}_2$  as a carbon source for growth;
- Their nitrogen comes from inorganic compounds such as  $\text{NH}_3$ ,  $\text{NO}_3^-$ , or  $\text{N}_2$ .
- Many autotrophs will **not grow on media** that contain organic matter
- This metabolic mode has also been called **chemotrophy**, **chemoautotrophy**, or **chemolithotrophy**.

# Bacterial nutrition

- In order to grow in nature or in the laboratory, a bacterium must have
  - an energy source,
  - a source of carbon
  - a permissive range of physical conditions such as O<sub>2</sub> concentration, Temp.and pH.
- The Major nutrients required by bacteria to grow includes
  - C, H, O, N, S. P, K, Mg, Fe, Ca, Mn, and traces of Zn, Co, Cu, and Mo.
- These elements are found in the form of
  - water,
  - inorganic ions,
  - small molecules,
  - macromolecules which serve either a structural or functional role in the cells.

# Carbon

- Organisms require a source of carbon for the synthesis of numerous organic compounds that comprise **protoplast**.
- required in the greatest amount
- Depending on carbon requirements, bacteria can be classified as
  - **Autotrophs**: Free-living, non-parasitic bacteria which use **carbon dioxide** as carbon source.
  - **Heterotrophs**: Parasitic bacteria require more complex organic compounds as their source of carbon and energy.



Human pathogenic bacteria are heterotrophs. Carbohydrates are carbon source

# Hydrogen and oxygen

- Obtained from water. essential for the growth and maintenance of cell.
- One of the most-prominent differences between bacteria is their requirement and response to atmospheric oxygen (O<sub>2</sub>)
  - Bacteria that require oxygen to grow are called obligate aerobic
  - Bacteria that grow only in the absence of oxygen obligate anaerobes
    - presence of oxygen actually poisons some of their key enzymes
  - microaerophilic or aerotolerant anaerobes grow better in low concentrations of oxygen
  - facultative anaerobes grow in the Presence or absence of O<sub>2</sub>
    - change their metabolic processes depending on the presence of oxygen
      - Respiration
      - fermentation



- The response of bacteria to oxygen is not determined simply by their **metabolic needs**.

### ❑ **Toxic Forms of Oxygen:**

- **Single Oxygen:** Extremely reactive form of oxygen, present in phagocytic cells.
- **Superoxide Free Radicals (O<sub>2</sub><sup>-</sup>):** Extremely toxic and reactive form of oxygen.
- organisms growing in atmospheric oxygen must produce an enzyme **superoxide dismutase (SOD)**, to get rid of them.

SOD is made by aerobes, facultative anaerobes, and aerotolerant anaerobes, **but not** by anaerobes or microaerophiles.

### **Reaction:**



**Hydrogen Peroxide (H<sub>2</sub>O<sub>2</sub>):** toxic and the active ingredient of several antimicrobials (e.g.: benzoyl peroxide).

- There are two different enzymes that break down hydrogen peroxide:

**A. Catalase:** Breaks hydrogen peroxide into water and O<sub>2</sub>.

Common. Produced by humans, as well as many bacteria.

### Catalase



**B. Peroxidase:** Converts hydrogen peroxide into water.

### Peroxidase



- **Sulfur:** Used to form **proteins** and some **vitamins** (thiamin and biotin).
- **Sources of sulfur:**
  - **Protein:** Most bacteria
  - **Hydrogen sulfide**
- **Sulfates:** Salts that dissociate to give  $\text{SO}_4^{2-}$
- **Phosphorus:** Used to form **DNA, RNA, ATP**, and **phospholipids**.  
**Sources:** Mainly **inorganic phosphate** salts and buffers.

# Temperature requirement

- Most **pathogenic** bacteria grow best at an optimum temperature of **37°C**.
- Based on temperature requirement, microorganisms can be broadly classified into
- **Psycrophilic**- which can grow in the range of **-5-20°C**
  - These bacteria include those which cause **spoilages of food** at refrigeration temperature (2-8°C).
- **Mesophilic**- which grow at **20-45°C** and show optimum growth at 37°C.
  - all medically important bacteria (pathogenic bacteria) belong to this group.

# Temperature requirement ...Cont'd

- **Thermophilic** – prefer high temperature (50-80°C)
  - May cause spoilage of under processed canned food
- **Hyperthermophilic**
  - grow at a temperature of >80°C
  - Some of them grow even at 250°C
  - are found in hot springs, geysers and industrial heated wastes

# pH requirement

Most pathogenic bacteria require a pH of **7.2-7.6** for their optimal growth.

Based on pH requirement bacteria can be classified as

- **Neutrophilic**:- bacteria grow best at neutral pH (pH=7)
  - Most pathogenic micro-organism best grow at neutral pH (pH=7)
- **Acidophilic**
  - Bacterial grow best at acidic pH (pH<7)
  - E.g. Lactobacilli, fungi and yeast
- **Alkalophilic**
  - Bacterial grow best at Alkaline pH (pH>7)
  - E.g. Vibrio cholerae grow at a pH of 8.6

# Salt

- **Halophiles**: Require moderate to large salt concentrations. Ocean water contains 3.5% salt.
  - Most bacteria in oceans.
- **Extreme or Obligate Halophiles**: Require very high salt concentrations (20 to 30%).
  - Bacteria in Dead Sea, brine vats.
- **Facultative Halophiles**: Do not require high salt concentrations for growth, but tolerate 2% salt or more.

## ➤ Growth factors.

- **essential** substances that the organism is **unable** to synthesize from available nutrients.
- required in **small amounts** by cells because they fulfill specific roles in biosynthesis.

## ➤ Growth factors are organized into three categories.

1. **purines and pyrimidines**: required for synthesis of nucleic acids (DNA and RNA)
2. **amino acids**: required for the synthesis of proteins
3. **vitamins**: needed as coenzymes and functional groups of certain enzymes

## ➤ Bacteria could be

- **Prototrophs**: Wild-type bacteria with normal growth requirements.
- **Auxotrophs**: Mutant bacteria, which require an additional growth factor



# BACTERIAL GROWTH

- It is an orderly increase in all the components of an organism.
- Bacterial growth involves **three complex** processes:
  - **metabolism, regulation, Cell division**
- The growth of a bacterial population occurs in a geometric or exponential manner:

**Generation time or population doubling time:** The interval of time between two cell division, or the time required for a bacterium to give rise to two daughter cells under optimum conditions

- Generation time varies considerably :
  - E. coli divides every 20 minutes.
  - Most bacteria divide every 1 to 3 hours.
  - Some bacteria require over 24 hours to divide.
- The generation time varies not only with the species but also with the amount of **nutrients, temperature, pH** and other environmental factors.

# Bacterial growth curve

- When bacteria are placed in a medium that provides all of the nutrients that are necessary for their growth
  - exhibits four phases of growth (**growth curve**)
- Represents the number of live cells in a **bacterial** population over a period of time.
- There are four distinct phases of the **growth curve**:
  - lag,
  - exponential (log),
  - stationary,
  - death

## Lag phase

- ✚ The period of adaptation with **active macro molecular synthesis**
  - ✚ DNA, RNA, various enzymes and other structural components, organisms grow in size only.
- ✚ It is the **preparation** time for reproduction;
- ✚ May last from one hour to several days

## Log Phase

- ✚ Cells **begin** to **divide** and **generation time** reaches a constant **minimum**.
- ✚ Period of most **rapid growth**.
- ✚ cell numbers increase in a logarithmic fashion  
**Number of cells produced > Number of cells dying**
- ✚ Cells are at **highest metabolic activity**.
- ✚ **most susceptible** to adverse environmental factors.
  - Radiation
  - Antibiotics

## Log (**logarithmic or exponential**) phase

- During this phase, the population can **double** approximately every **30 minutes** with fast growing bacteria
- It has limited duration because of:-
  - Exhaustion of nutrients
  - Accumulation of toxic metabolic end products
  - Rise in cell density
  - Change in pH and
  - Decrease in oxygen tension (in case of aerobic organisms)

## Stationary Phase:

Occur when **nutrients depletion** or **toxic products** cause growth to slow until the number of new cells produced balances the number of cells that die resulting in a steady state

- Population size begins to stabilize.
- Overall cell number does not increase.
- Cell division begins to slow down.

**Number of cells produced = Number of cells dying**

**Factors that slow down microbial growth:**

- ✓ Accumulation of toxic waste materials
- ✓ Acidic pH of media
- ✓ Limited nutrients
- ✓ Insufficient oxygen supply

## The death/decline phase

Due to **depletion of nutrients** and accumulation of toxic end products the number of bacteria dying is much more than those dividing and hence there is gradual decline in the total number of organism.

- There is drastic decline in viable cells
- Population size begins to decrease.
- Cell number decreases at a logarithmic rate.
- Cells lose their ability to divide.
- A few cells may remain alive for a long period of time.

**Number of cells dying > Number of cells produced**

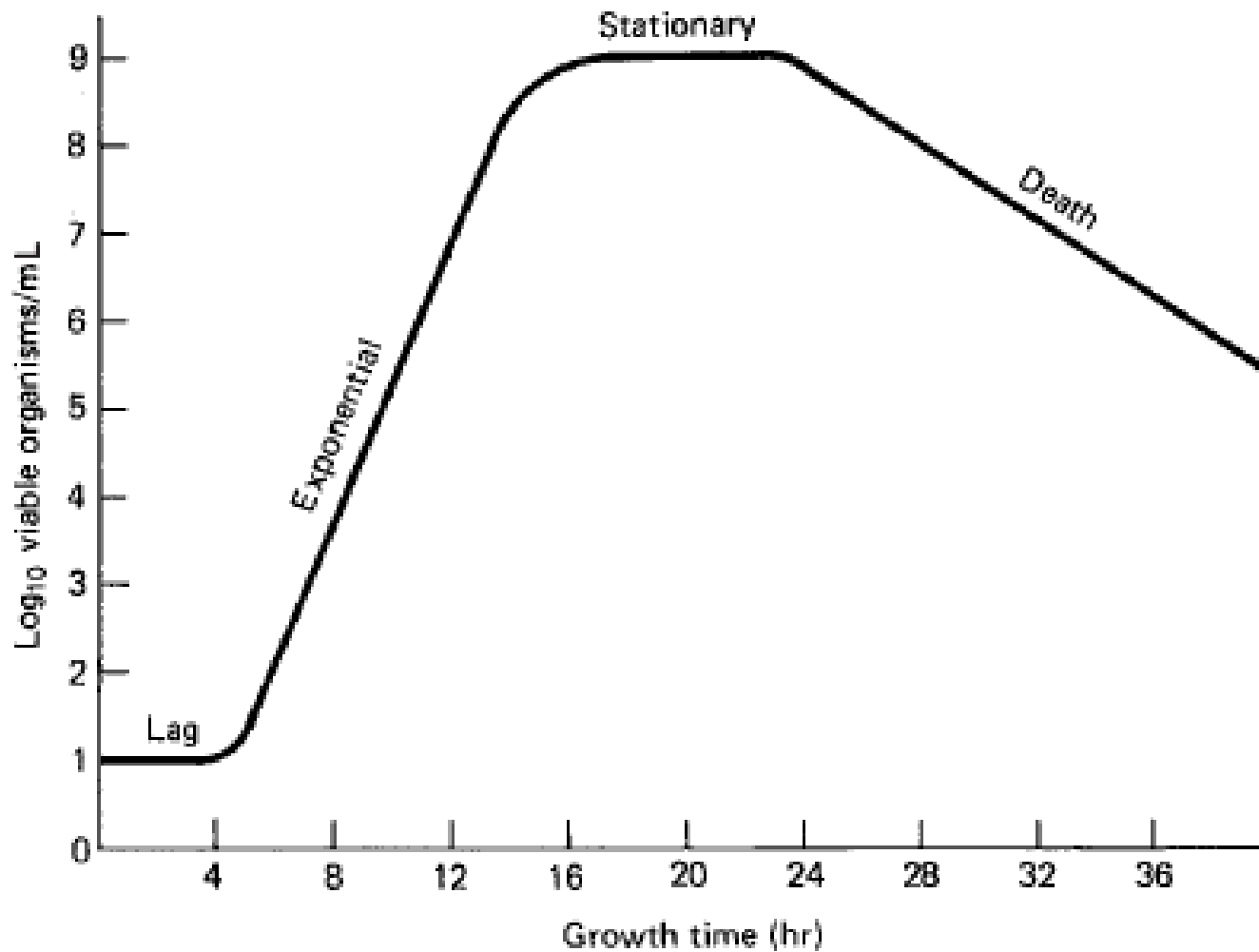


Fig. Bacterial growth curve

# Measuring Microbial Growth

## ❑ Direct cell counts:-

- counting chambers(Petroff Hauser )
- electronic counters
- plating methods (spread, pour plate)
- membrane filtration methods

## ❑ Indirect Methods

- Turbidity
- Metabolic Activity



## Counting chambers

- easy, inexpensive, and quick, useful for counting both eucaryotes and procaryotes

cannot distinguish living from dead cells

## Electronic counters

- Such as Coulter counter microbial suspension forced through small hole or orifice
- movement of microbe through orifice impacts electric current that flows through orifice
- quick and easy to use, useful for large microorganisms and blood cells, but not prokaryotes

cannot distinguish living from dead cells

## membrane filters

- cells filtered through **special membrane** that provides **dark background** for observing cells
- cells are stained with **fluorescent dyes**
- useful for counting bacteria
- with certain dyes, can distinguish living from dead cells